
MA 578: BAYESIAN STATISTICS

Chapter 6: Bayesian Linear Regression

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1 Introduction

In many applications, our goal is to understand the statistical relationship between a primary variable of interest y and a set of factors $\mathbf{x} = (x_1, \dots, x_p)$.

- Example: Which factors most strongly influence how long it takes a freshly graduated student to find a job?

The primary variable of interest y is called the **response** or **outcome variable**. The variables $\mathbf{x} = (x_1, \dots, x_p)$ are called **explanatory variables**, **factors**, or **covariates**.

Statistically, such problems boil down to modeling the conditional distribution $p(y \mid \mathbf{x})$ of y given $\mathbf{x}_{1:p}$. A **regression model** postulates a parametric or semiparametric form for this conditional distribution.

Data.

- Outcomes: $y_1, \dots, y_n$
- Covariates: $\mathbf{x}_1, \dots, \mathbf{x}_n$, where $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$

We begin with the case where y is continuous. For this setting, we study the **linear regression model**. Later in the course, we will extend these ideas to **generalized linear models (GLMs)**, which are suitable when y is not continuous.

2 The Linear Model

The linear model is a fundamental and versatile tool for data analysis. It is used for

- parameter estimation,
- prediction,
- data description.

We first review the frequentist **ordinary least squares (OLS)** estimator, then introduce the **Bayesian** approach and compare the two.

How should we estimate $p(y \mid \mathbf{x})$?

A naive approach would be to estimate the distribution for each possible combination of covariates $\mathbf{x}$. However, in most real-world datasets, there are not enough observations for such a fully nonparametric approach.

An alternative is to assume a **smooth model** for $p(y \mid \mathbf{x})$. The key advantage is that data at one value of $\mathbf{x}$ provide information about nearby values of $\mathbf{x}$.

The simplest smooth model of practical interest is the **linear regression model**, which assumes that

$$\mathbb{E}[Y \mid \mathbf{x}] = \beta_1 x_1 + \cdots + \beta_p x_p = \mathbf{x}^\top \beta,$$

where $\beta = (\beta_1, \dots, \beta_p)^\top$.

Remark. The vector $\mathbf{x}$ can include not only the raw covariates but also transformations or interactions of them—such as squared terms x_j^2 or products $x_j x_k$ —allowing the model to capture certain nonlinear relationships.

From $\mathbb{E}[Y \mid \mathbf{x}]$ to $p(y \mid \mathbf{x})$: The Normal Linear Regression Model

We assume that the explanatory variables are nonrandom (fixed design). Thus, the randomness in the observed data $(y_i, \mathbf{x}_i)$ arises only from the response variable:

$$y_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n$ are independent with common distribution $\mathcal{N}(0, \sigma^2)$.

Under this model,

$$\begin{aligned} p(y_1, \dots, y_n \mid \mathbf{x}_1, \dots, \mathbf{x}_n, \beta, \sigma^2) &= \prod_{i=1}^n p(y_i \mid \mathbf{x}_i, \beta, \sigma^2) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2\right). \end{aligned}$$

Matrix form. Define

$$\begin{aligned} y &= \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \\ X &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}_{n \times p}. \end{aligned}$$

Then

$$y = X\beta + \epsilon.$$

The conditional distribution is

$$p(y \mid X, \beta, \sigma^2) = \mathcal{N}(y \mid X\beta, \sigma^2 I_n) = \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2\right),$$

where $\|v\|_2 = \sqrt{\sum_i v_i^2}$ denotes the Euclidean norm.

Maximum likelihood estimation. The MLE of β maximizes $p(y \mid X, \beta, \sigma^2)$, equivalently minimizing $\|y - X\beta\|_2^2$.

Claim. The maximum likelihood (and least squares) estimator of β is

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

Proof.

$$\begin{aligned} \|y - X\beta\|_2^2 &= y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta, \\ \frac{\partial}{\partial \beta} \|y - X\beta\|_2^2 &= -2X^\top y + 2X^\top X \beta, \end{aligned}$$

$$\text{Setting derivative to zero: } X^\top X \hat{\beta} = X^\top y.$$

$$\text{Hence } \hat{\beta} = (X^\top X)^{-1} X^\top y.$$

□

This estimator $\hat{\beta} = \hat{\beta}_{\text{OLS}}$ is called the **ordinary least squares (OLS)** estimator.

Sampling distribution. Because $y = X\beta + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, we have

$$\hat{\beta} = (X^\top X)^{-1} X^\top y = \beta + (X^\top X)^{-1} X^\top \epsilon,$$

therefore,

$$\hat{\beta} \sim \mathcal{N}\left(\beta, \sigma^2 (X^\top X)^{-1}\right).$$

Estimating σ^2 . An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n-p} \|y - X\hat{\beta}\|_2^2.$$

The denominator $n-p$ is the residual degrees of freedom.

Here, $y - X\hat{\beta}$ is the vector of residuals, and $\text{SSR}(\hat{\beta}) = \|y - X\hat{\beta}\|_2^2$ is the **sum of squared residuals**.

Inference for individual coefficients. Let $\Omega = (X^\top X)^{-1}$ and denote its i -th diagonal element by Ω_{ii} . Then, for each coefficient β_i ,

$$T_i = \frac{\hat{\beta}_i - \beta_i}{\sqrt{\hat{\sigma}^2 \Omega_{ii}}} \sim \mathsf{T}_{n-p},$$

which can be used to test

$$H_0 : \beta_i = 0 \quad \text{versus} \quad H_1 : \beta_i \neq 0,$$

with two-sided p -value

$$p_i = \Pr(|T_i| > |t_i|).$$

Example (oxygen uptake)

Goal. Study how two exercise programs (treatments) affect the change in maximum oxygen uptake (outcome).

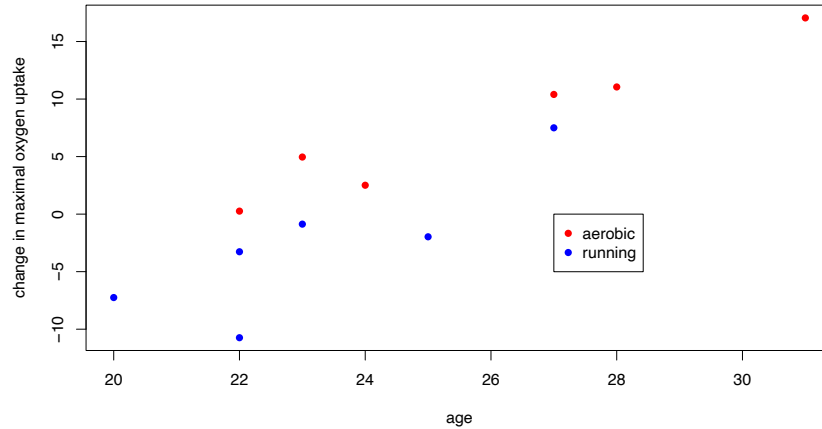
- **Treatment 1:** 12-week running program.
- **Treatment 2:** 12-week aerobic program.

Data. 6 subjects were assigned to Treatment 1, and 6 to Treatment 2. The outcome is the pre-post treatment change in maximum oxygen uptake.

We notice that the covariate “age” can modify the outcome. Therefore, we ask the following question:

Question. How does a subject’s outcome depend on treatment, adjusting for age?

Therefore, our goal is to estimate the conditional distribution of the outcome given treatment and age.



Normal linear regression with interaction. Let $T \in \{0, 1\}$ indicate treatment assignment ($T = 1$ for Treatment 1; $T = 0$ for Treatment 2), and let **age** denote age. We use the model

$$Y = \beta_1 \cdot 1 + \beta_2 T + \beta_3 \text{age} + \beta_4 (\text{age} \times T) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

Equivalently, with regressors $\mathbf{x} = (x_1, x_2, x_3, x_4) = (1, T, \text{age}, \text{age} \cdot T)$,

$$\mathbb{E}[Y \mid \mathbf{x}] = \mathbb{E}[Y \mid \text{age}, T] = \begin{cases} \beta_1 + \beta_3 \text{age}, & T = 0 \text{ (Treatment 2)}, \\ (\beta_1 + \beta_2) + (\beta_3 + \beta_4) \text{age}, & T = 1 \text{ (Treatment 1)}. \end{cases}$$

Interpretation. β_2 is the treatment effect on the intercept (difference at age = 0); β_4 is the treatment-age interaction (difference in slopes).

If we impose $\beta_2 = \beta_4 = 0$, the model ignores treatment and fits a single line in age for both groups.

Design matrix and response. With $n = 12$ and columns ordered as $(1, T, \text{age}, \text{age} \cdot T)$,

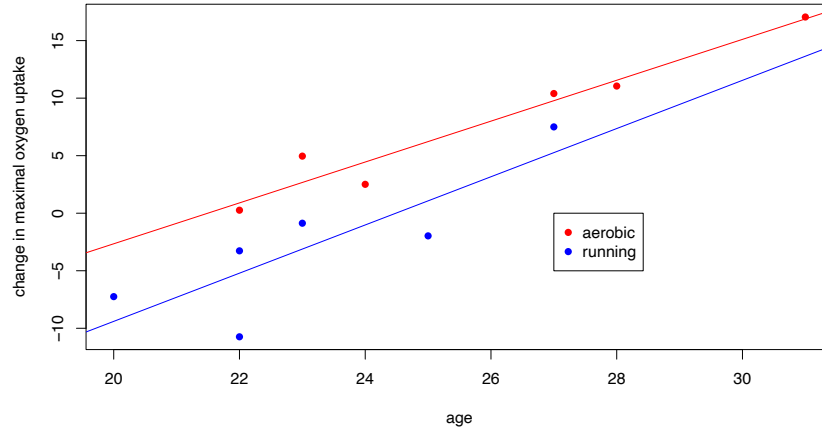
$$\begin{aligned} x_{1:12,1} &= (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1), \\ x_{1:12,2} &= (1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0), \\ x_{1:12,3} &= (23, 22, 22, 25, 27, 20, 31, 23, 27, 28, 22, 24), \\ x_{1:12,4} &= (23, 22, 22, 25, 27, 20, 0, 0, 0, 0, 0, 0), \\ y &= (-0.87, -10.74, -3.27, -1.97, 7.50, -7.25, 17.05, 4.96, 10.40, 11.05, 0.26, 2.51). \end{aligned}$$

Then

$$X^\top X = \begin{bmatrix} 12 & 6 & 294 & 155 \\ 6 & 6 & 155 & 155 \\ 294 & 155 & 7314 & 4063 \\ 155 & 155 & 4063 & 4063 \end{bmatrix}, \quad X^\top y = \begin{bmatrix} 29.63 \\ 46.23 \\ 978.81 \\ 1298.79 \end{bmatrix}, \quad \Rightarrow \quad \hat{\beta}_{\text{OLS}} = \begin{bmatrix} -51.29 \\ 13.11 \\ 2.09 \\ -0.32 \end{bmatrix}.$$

Fitted lines by treatment.

- **Treatment 2** ($T = 0$): intercept = -51.29 , slope = 2.09 .
- **Treatment 1** ($T = 1$): intercept = $-51.29 + 13.11$, slope = $2.09 - 0.32$.



Residual variance and standard errors. With $p = 4$ regressors (including the intercept), the residual degrees of freedom are $n - p = 8$. The OLS estimate of σ^2 is

$$\hat{\sigma}_{\text{OLS}}^2 = \frac{\|y - X\hat{\beta}\|_2^2}{n - p} = 8.54.$$

Using estimate of the sampling variance $\widehat{\text{var}}(\hat{\beta}) = \hat{\sigma}_{\text{OLS}}^2 (X^\top X)^{-1}$, the standard errors for $(\beta_1, \beta_2, \beta_3, \beta_4)$ are

$$\text{SE}(\hat{\beta}) = (12.25, 15.76, 0.53, 0.65).$$

Reporting. For each coefficient, the usual t -statistic with $n - p = 8$ d.f. is

$$t_i = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}, \quad i = 1, \dots, 4,$$

with two-sided p -value $\Pr(|T_8| > |t_i|)$.

3 Bayesian Linear Regression Model

In the Bayesian framework, inference proceeds by combining the likelihood function with a prior distribution to obtain the posterior. Specifically, with fixed design matrix X , we write

$$p(\beta, \sigma^2 \mid y, X) \propto p(y \mid X, \beta, \sigma^2) p(\beta, \sigma^2 \mid X).$$

Since X is assumed fixed, it can also appear in the specification of the prior.

Two common choices for priors in this model are:

- Zellner's g -prior,
- The normal-inverse-gamma conjugate prior.

We focus on Zellner's g -prior in this section and return to the general normal-inverse-gamma prior in the next chapter.

3.1 Sampling Model and General Normal Prior

Recall the sum of squared residuals:

$$\text{SSR}(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 = (y - X\beta)^\top (y - X\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Hence the sampling model is

$$\begin{aligned} p(y \mid X, \beta, \sigma^2) &= \mathcal{N}(y \mid X\beta, \sigma^2 I_n) \\ &\propto \exp \left\{ -\frac{1}{2\sigma^2} \text{SSR}(\beta) \right\} \\ &= \exp \left[-\frac{1}{2\sigma^2} (y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta) \right]. \end{aligned}$$

Claim. The conjugate prior for β (conditional on σ^2) is a multivariate normal distribution.

Proof. Assume

$$\beta \mid X, \sigma^2 \sim \mathcal{N}(\beta_0, \Sigma_0).$$

Then, up to a proportionality constant,

$$\begin{aligned} p(\beta \mid y, X, \sigma^2) &\propto p(y \mid X, \beta, \sigma^2) p(\beta \mid X, \sigma^2) \\ &\propto \exp \left[-\frac{1}{2} \left(\beta^\top (X^\top X / \sigma^2 + \Sigma_0^{-1}) \beta - 2\beta^\top (X^\top y / \sigma^2 + \Sigma_0^{-1} \beta_0) \right) \right]. \end{aligned}$$

This has the same shape as a multivariate normal distribution with

$$\begin{aligned} \Sigma_n &= \text{var}(\beta \mid y, X, \sigma^2) = (\Sigma_0^{-1} + X^\top X / \sigma^2)^{-1}, \\ \beta_n &= \mathbb{E}[\beta \mid y, X, \sigma^2] = \Sigma_n (\Sigma_0^{-1} \beta_0 + X^\top y / \sigma^2). \end{aligned}$$

□

We see that we have the same form for the posterior hyper parameters as those we observed in Chapter 5 in studying multivariate normal models: the posterior precision equals the sum of the prior precision and the data precision; the posterior mean is a weighted (precision-weighted) average of the prior mean and the OLS estimate.

Limiting cases.

- If $\Sigma_0^{-1} \rightarrow 0$ (i.e., an uninformative or “flat” prior), then the posterior mean estimator (which since we are dealing with a normal distribution, is also the MAP estimator) will be the same as MLE estimator (i.e., OLS estimator). That is,

$$\beta_n = \hat{\beta}_{\text{MAP}} = \hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{OLS}}.$$

- If $\beta_0 = 0$ and $\Sigma_0 = (\sigma^2/\alpha)I_p$, then

$$\beta_n = \hat{\beta}_{\text{MAP}} = \hat{\beta}_{\text{ridge}} = (X^\top X + \alpha I_p)^{-1} X^\top y,$$

which coincides with the **ridge regression** estimator. Thus, ridge regression can be viewed as a MAP estimate under a normal prior centered at zero.

3.2 Zellner’s g -Prior

In this approach, we use the following specific choice of the hyper parameters of the prior distribution that reflects the geometry of the design matrix X :

$$\beta_0 = 0, \quad \Sigma_0 = k(X^\top X)^{-1},$$

where $k > 0$ is a scalar. A popular and scale-consistent choice is $k = g\sigma^2$, for some positive value g , giving the g -prior:

$$\beta \mid \sigma^2, X \sim \mathcal{N}(0, g\sigma^2(X^\top X)^{-1}).$$

Posterior under the g -prior. Under this choice, $p(\beta \mid y, X, \sigma^2)$ will be multivariate normal distribution with

$$\text{var}(\beta \mid y, X, \sigma^2) = \left(\frac{X^\top X}{g\sigma^2} + \frac{X^\top X}{\sigma^2} \right)^{-1} = \frac{g}{g+1} \sigma^2 (X^\top X)^{-1},$$

$$\mathbb{E}[\beta \mid y, X, \sigma^2] = \Sigma_n(\Sigma_0^{-1}\beta_0 + X^\top y/\sigma^2) = \frac{g}{g+1} (X^\top X)^{-1} X^\top y = \frac{g}{g+1} \hat{\beta}_{\text{OLS}}.$$

Hence, the posterior mean “shrinks” the OLS estimator toward zero by a factor $g/(g+1)$: large g implies weak shrinkage (prior less influential), while small g implies stronger regularization.

Remark (Equivalent prior sample size for Zellner’s g -prior). Under Zellner’s g -prior, the posterior precision adds as

$$\underbrace{\frac{1}{\sigma^2} X^\top X}_{\text{data precision}} + \underbrace{\frac{1}{g\sigma^2} X^\top X}_{\text{prior precision}}.$$

Hence the prior precision equals a fraction $1/g$ of the data precision. Because information in the normal linear model scales linearly with n , this can be read as an *equivalent prior sample size*

$$n_{\text{prior}} = \frac{n}{g}.$$

In particular:

- $g = n$ yields the *unit-information prior*: the prior contributes about as much information as one typical observation.
- $g < n$: the prior is stronger than one observation ($n_{\text{prior}} > 1$).
- $g > n$: the prior is weaker than one observation ($n_{\text{prior}} < 1$).

Advantages of the g -Prior

1. Scale Invariance. Estimates are invariant to changes in the units or scaling of the regressors. For example,

Option 1: Use covariate $\tilde{x}_3$: age in months $\rightarrow \tilde{\beta}_3$

Option 2: Use covariate x_3 : age in years $\rightarrow \beta_3$

We expect that posterior dist. for $12\tilde{\beta}_3$ in model with $\tilde{x}_3$ = posterior dist. for β_3 in model with x_3 . That is, whether age is measured in years or months, the posterior for the corresponding coefficient (properly rescaled) remains the same.

Principle of invariance:

Consider regressors X and $\tilde{X} = XH$, for some $p \times p$ matrix H

β is obtained from y and X

$\tilde{\beta}$ is obtained from y and $\tilde{X}$

The posterior distribution for β should be the same as the posterior distribution for $H\tilde{\beta}$. If so, then the posterior distribution for $X\beta$ will be the same as the posterior distribution for $\tilde{X}\tilde{\beta}$.

This desirable invariance holds under the g -prior.

Proof.

$$\begin{aligned}\tilde{X}\tilde{\beta} &= \tilde{X}(\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top y \\ &= XH(H^\top X^\top XH)^{-1} H^\top X^\top y \\ &= XHH^{-1}(X^\top X)^{-1} H^{-\top} H^\top X^\top y \\ &= X(X^\top X)^{-1} X^\top y \\ &= X\beta.\end{aligned}$$

□

2. Conjugacy with inverse-gamma prior on σ^2 . If the prior for σ^2 is

$$p(\sigma^2 | X) = \text{IG}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right),$$

then the posterior is

$$\sigma^2 | y, X \sim \text{IG}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + \text{SSR}_g}{2}\right),$$

where

$$\text{SSR}_g = y^\top \left(I - \frac{g}{g+1} X(X^\top X)^{-1} X^\top \right) y.$$

Therefore, we can sample from the joint posterior distribution $p(\sigma^2, \beta | y, X)$ by sampling from $p(\sigma^2 | y, X)$ and $p(\beta | \sigma^2, y, X)$, and no MCMC approximation is required:

Sampling from the posterior $p(\sigma^2, \beta | y, X)$.

1. Draw $\sigma^2 \sim \text{IG}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + \text{SSR}_g}{2}\right)$;
2. Draw $\beta \sim \mathcal{N}\left(\frac{g}{g+1} \hat{\beta}_{\text{OLS}}, \frac{g}{g+1} \sigma^2 (X^\top X)^{-1}\right)$.

Remarks

1. The matrix X must be full rank for $(X^\top X)^{-1}$ to exist. If X is nearly singular, a ridge correction such as $X^\top X + \varepsilon I_p$ can be used.
2. Although the g -prior depends on X , this does not violate the “no data in prior” rule because X is treated as fixed (non-random). Still, this connection makes the g -prior conceptually similar to an *empirical Bayes* prior, which we will study later.

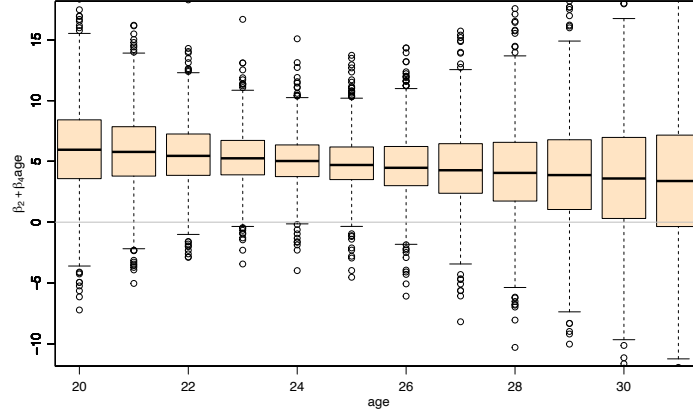


Figure 1: Posterior mean of the average treatment effect $\mathbb{E}[Y \mid T = 1, \text{age}] - \mathbb{E}[Y \mid T = 0, \text{age}]$ under the g -prior.

3. When prior information about σ^2 is unavailable, a common “non-informative” choice is

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2}.$$

A non-informative prior is a prior distribution that perturbs the least the likelihood. However these priors typically fail to integrate to one, hence are not really densities. We call them *improper densities*. Hence, the use of non-informative priors requires the extra step of checking that the posterior distribution is well-defined.

Back to the Oxygen Uptake Example

- **Prior specification.** We adopt the invariant g -prior with the following hyperparameters:

$$g = n = 12, \quad \nu_0 = 1, \quad \sigma_0^2 = \hat{\sigma}_{\text{OLS}}^2 = 8.54.$$

- **Posterior mean.** Using the g -prior result $\mathbb{E}[\beta \mid y, X, \sigma^2] = \frac{g}{g+1} \hat{\beta}_{\text{OLS}}$, we obtain

$$\mathbb{E}[\beta \mid y, X, \sigma^2] = \frac{12}{13} \begin{bmatrix} -51.29 \\ 13.11 \\ 2.09 \\ -0.32 \end{bmatrix} = \begin{bmatrix} -47.36 \\ 12.10 \\ 1.93 \\ -0.30 \end{bmatrix}.$$

- **Posterior uncertainty.** Based on 1000 independent Monte Carlo samples from the posterior, the posterior standard deviations (approximate posterior standard errors) are

$$[14.41, 18.62, 0.62, 0.77]^\top.$$

Compare these to the corresponding OLS standard errors $[12.25, 15.76, 0.53, 0.65]^\top$; the Bayesian estimates show slightly higher uncertainty due to the prior regularization.

- **Posterior treatment effect.** The average treatment effect (ATE) at age `age` is given by

$$\mathbb{E}[Y \mid T = 1, \text{age}] - \mathbb{E}[Y \mid T = 0, \text{age}] = \beta_2 + \beta_4 \text{age}.$$

There seems to be strong evidence of a difference at younger ages and less evidence at the older ages.